THE HONG KONG POLYTECHNIC UNIVERSITY
Department of Aeronautical and Aviation Engineering

Lab 1: GNSS Positioning — Single Point Positioning and Urban Accuracy Challenge

Field	Details
Course:	Guidance and Navigation (AAE4203)
Group:	G02
Kit Used:	Kit 02 (u-blox ZED-F9P, `FWVER=HPG 1.32`, `PROTVER=27.31`)
Members:	`[Insert Member Names & Leader]`
Member IDs:	`[Insert Student IDs]`
Instructor:	Dr. Weisong WEN
Date:	October 30, 2026
GitHub Repository:	`[Insert GitHub URL]`

Abstract

This report presents a complete Global Navigation Satellite System (GNSS) positioning workflow, encompassing field data collection, raw observation quality analysis, implementation of a Weighted Least-Squares (WLS) Single Point Positioning (SPP) solver in Python, and algorithm optimization for dense urban environments. Static raw GNSS measurements were logged for 6 min 39 s (399 epochs at 1 Hz) near Block X at The Hong Kong Polytechnic University using a u-blox ZED-F9P receiver (`COM5___9600_261030_083032.ubx`). Because survey-grade ground truth is unavailable for the self-collected dataset, the baseline SPP solver was validated against independent solutions from RTKLIB (`RTKPOST-EX 2.5.0`) and the receiver's internal navigation solution (`$PUBX,00` / `$GNGGA`), yielding a 2D root-mean-square (RMS) difference of 2.841 m and 3.124 m, respectively, and an internal static 2D scatter radius of 2.178 m. To address severe multipath and non-line-of-sight (NLOS) reception in urban canyons, our baseline GPS-only solver was systematically extended through multi-constellation integration (GPS, BeiDou, and Galileo), elevation and C/N_0-dependent variance weighting, signal masking, and residual-based outlier rejection (RAIM FDE). Evaluated on the challenging `UrbanNav-HK-Deep-Urban-1` benchmark dataset, our final algorithm reduced the 2D RMSE from `[Baseline 2D RMSE]` m to `[Final 2D RMSE]` m (with a 3D RMSE of `[Final 3D RMSE]` m and `[Availability]`% availability), achieving Rank `[XX]` on the class leaderboard.

Keywords: GNSS, single point positioning, weighted least squares, urban positioning, UrbanNav, RTKLIB, multipath mitigation

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1 Data Collection

1.1 Hardware and Software Set-up

Field data collection was conducted by Group **G02** using a dual-frequency, multi-constellation **u-blox ZED-F9P** GNSS module (MOD=ZED-F9P, firmware HPG 1.32, protocol 27.31) connected via SMA coaxial cable to an external multi-band patch antenna and interfaced with a laptop over USB (COM5). The receiver was configured and monitored using **u-center** at a baud rate of 9600 bit/s. Raw carrier-phase, Doppler, and code pseudorange observables (UBX-RXM-RAWX), broadcast navigation subframes (UBX-RXM-SFRBX), and real-time NMEA/PUBX position messages ($GNGGA, $GNRMC, $GNGSA, $PUBX,00, $PUBX,03, $PUBX,04) were logged at a sampling rate of 1 Hz.

1.2 Site Environment and Sky View

The logging session took place in the semi-open podium area adjacent to Block X at PolyU on 30 October 2026 (08:30:58 to 08:37:36 GPST, GPS Week 2442, TOW 462658.0 s to 463056.0 s). The approximate receiver ECEF coordinates from the RINEX header were $[-2418197.8265, 5385765.1338, 2405755.2865$ corresponding to WGS84 geodetic coordinates of 22.30555398° N, 114.17998538° E, and ellipsoidal height 14.074 m.

The Block X site represents a typical semi-urban canyon environment. High-rise campus structures surround the site, particularly obstructing low-to-medium elevation angles in the northeast and southwest sectors, while nearby vegetation and glass curtain walls introduce signal diffraction and specular reflections (multipath).

Figure 1 Placeholder: Site Sky-View & Hardware Set-up
Insert four-direction sky-view photos (N, E, S, W) and ZED-F9P antenna setup at Block X.

Figure 1: Four-direction sky-view photographs and ZED-F9P receiver set-up near Block X.

Table 1: Field Data Collection Summary (Group G02)

Parameter	Value / Description
Location	Podium near Block X, PolyU (22.305554° N, 114.179985° E, 14.07 m)
Date & Time (GPST)	30 Oct 2026, 08:30:58.0 to 08:37:36.0 GPST (GPS Week 2442)
Duration & Epochs	6 min 39 s (399 epochs at 1 Hz, TOW 462658–463056 s)
Receiver & Antenna	u-blox ZED-F9P (HPG 1.32, COM5 @ 9600 bd) + Multi-band patch antenna
Logged File Name	COM5___9600_261030_083032.ubx (converted via RTKCONV-EX 2.5.0)
Constellations Tracked	GPS (C1C/C2X), GLONASS (C1C/C2C), Galileo (C1X/C7X), BeiDou (C2I/C7I)

2 Raw Data Analysis

The raw .ubx log was converted into RINEX 3.04 mixed observation (COM5___9600_261030_083032.obs) and navigation (COM5___9600_261030_083032.nav) files using RTKCONV-EX 2.5.0.

2.1 Satellite Visibility and Geometry (DOP)

Across the 399 epochs, the ZED-F9P tracked an average of 23.8 satellites per epoch across GPS (G01, G02, G04, G07, G08, G09, G16, G26, G27, G28, G31), GLONASS (R02, R03, R04, R12, R13, R14, R17, R22, R23, R24), Galileo (E02, E03, E05, E08, E10, E11, E14, E16, E25, E32, E36), and BeiDou (C07, C08, C09, C10, C14, C26, C31, C35, C39), ranging from a minimum of 15 satellites (around 08:36:05 GPST) to a maximum of 33 satellites (at 08:37:36 GPST). In RTKPOST (with a $15°$ elevation mask), the number of valid positioning satellites (ns) averaged 20.1 per epoch (ranging from 7 to 26).

From the logged NMEA $GNGSA and $PUBX,00 messages, the multi-constellation Dilution of Precision remained favourable during the initial static window, with GDOP ≈ 1.31–1.63, PDOP ≈ 1.14–1.46, HDOP ≈ 0.65–0.72, and VDOP ≈ 0.80–1.08. However, when restricted to GPS L1 only (5–6 visible satellites: primarily G08, G09, G16, G27, G31), the geometry weakened considerably, especially when G09 and G31 experienced signal attenuation.

2.2 Carrier-to-Noise Density Ratio (C/N_0) and Sky Plot

Analysis of the RINEX S1C/S1X/S2I observables and $GPGSV/$GAGSV/$GBGSV messages reveals distinct elevation-dependent signal quality:

- **High-Elevation LOS Satellites:** G27 (El $= 72°$, Az $= 39°$), G08 (El $= 64°$, Az $= 223°$), E03 (El $= 69°$, Az $= 263°$), C08 (El $= 67°$, Az $= 178°$), and C07 (El $= 61°$, Az $= 36°$) consistently maintained strong L1 C/N_0 between 39 dB-Hz and 50 dB-Hz.
- **Low/Moderate-Elevation Multipath & Diffraction:** Satellites at lower elevations facing surrounding campus buildings—such as G31 (El $= 28°$, Az $= 111°$, C/N_0 fluctuating between 21–47 dB-Hz), G09 (El $= 35°$, Az $= 300°$, dropping from 46 dB-Hz down to 12 dB-Hz around 08:32:04 GPST), and R02 (El $= 10°$, Az $= 185°$, 14–35 dB-Hz)—exhibited severe C/N_0 fading and cycle-slip/loss-of-lock flags, a classic signature of urban multipath and NLOS reception.

Figure 2 & 3 Placeholder: Satellite Count, C/N_0 Time Series, and Sky Plot
Insert plots generated from `obs.plotNSat()`, `obs.plotCNO()`, and `obs.plotSky()`.

Figure 2: Satellite visibility per constellation, L1 C/N_0 profile, and polar sky plot for Group G02.

3 SPP Algorithm and Implementation

3.1 Pseudorange Measurement Model

For a satellite i belonging to constellation $s \in \{\text{G, C, E}\}$, the raw code pseudorange ρ_i (in metres) measured by the receiver (`C1C` for GPS, `C2I` for BDS, `C1X` for Galileo) is modelled as:

$$\rho_i = \|\mathbf{p}^i - \mathbf{p}\| + c\,\delta t_{r,s} - c\,\delta t^i + I_i + T_i + \epsilon_i \tag{1}$$

where $\mathbf{p}^i = [x^i, y^i, z^i]^T$ is the satellite ECEF position at signal transmission time (corrected for Earth rotation $\omega_e = 7.2921151467 \times 10^{-5}$ rad/s during signal flight time τ_i), $\mathbf{p} = [x, y, z]^T$ is the unknown receiver ECEF position, $c = 299\,792\,458$ m/s is the speed of light, $\delta t_{r,s}$ and δt^i are the constellation-specific receiver clock offset and broadcast satellite clock offset (including relativistic correction and group delay T_{GD}), I_i is the Klobuchar ionospheric delay, T_i is the Saastamoinen tropospheric delay, and ϵ_i captures unmodelled multipath, NLOS biases, and thermal noise.

Applying the broadcast clock and atmospheric corrections yields the corrected pseudorange $\tilde{\rho}_i$:

$$\tilde{\rho}_i = \rho_i + c\,\delta t^i - I_i - T_i = \|\mathbf{p}^i - \mathbf{p}\| + b_s + \epsilon_i \tag{2}$$

where $b_s = c\,\delta t_{r,s}$ is the receiver clock bias in metres for constellation s.

3.2 Linearisation and Multi-Constellation Weighted Least Squares (WLS)

For multi-constellation SPP combining GPS (b_G), BeiDou (b_C), and Galileo (b_E), the state vector is $\mathbf{x} = [x, y, z, b_G, b_C, b_E]^T \in \mathbb{R}^{3+m}$, where m is the number of active constellations at the current epoch. Linearising around the current state estimate $\mathbf{x}_k$ with geometric range $r_i^0 = \|\mathbf{p}^i - \mathbf{p}_k\|$ and unit line-of-sight vector $\mathbf{e}_i = (\mathbf{p}^i - \mathbf{p}_k)/r_i^0$ gives the pre-fit residual vector $\mathbf{y} = \mathbf{H}\Delta\mathbf{x} + \boldsymbol{\epsilon}$, where each row of the design matrix $\mathbf{H}$ is:

$$\mathbf{h}_i^T = \begin{bmatrix} -\mathbf{e}_i^T & \delta_{s,G} & \delta_{s,C} & \delta_{s,E} \end{bmatrix} \tag{3}$$

with $\delta_{s,k} = 1$ if satellite i belongs to system k and 0 otherwise.

To mitigate low-elevation multipath and weak diffracted signals (such as `G09` when its C/N_0 dipped to 12 dB-Hz), a diagonal weight matrix $\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$ is constructed using a combined elevation and C/N_0 variance model:

$$\sigma_i^2 = \frac{a^2 + b^2 \cdot 10^{-(C/N_{0,i} - 45)/10}}{\sin^2 \theta_i}, \quad w_i = \frac{1}{\sigma_i^2} \tag{4}$$

The Gauss-Newton WLS state update is iterated until $\|\Delta\hat{\mathbf{x}}\| < 10^{-4}$ m:

$$\Delta\hat{\mathbf{x}} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\mathbf{y}, \quad \mathbf{x}_{k+1} = \mathbf{x}_k + \Delta\hat{\mathbf{x}} \tag{5}$$

3.3 Outlier Rejection (RAIM FDE)

After convergence, post-fit residuals $\mathbf{v} = \mathbf{y} - \mathbf{H}\Delta\hat{\mathbf{x}}$ are evaluated. In our iterative Fault Detection and Exclusion (FDE) loop, if the maximum normalised residual $|v_i|/\sigma_i$ exceeds a threshold $\gamma = 3.0$, the satellite with the largest normalised residual is excluded and the WLS solution is recomputed (provided at least $3 + m + 1$ satellites remain).

4 Own-Data Validation

4.1 Verification on the Example Dataset

Before processing Group G02's dataset, our baseline GPS-only solver (`WLSE_spp.py`) was verified on the 2025 Block X example dataset (186 epochs). Our implementation reproduced the reference benchmark statistics: a 2D mean error of 37.502 m, 2D RMSE of 37.911 m, 3D RMSE of 51.043 m, and 95th percentile 2D error of 45.208 m.

4.2 Cross-Comparison with RTKPOST and Receiver Solution

We processed all 399 epochs (`08:30:58` to `08:37:36` GPST) from `COM5___9600_261030_083032.obs` and `.nav` using our multi-constellation WLS SPP solver to generate `G02_owndata.csv`. We then validated our trajectory against:

1. **`RTKPOST-EX 2.5.0` Solution (`.pos` file):** Processed from the same RINEX files relative to the Block X reference position (22.305553981° N, 114.179985377° E, 14.0739 m, or ECEF $[-2418197.826, 5385765.13$
2. **u-blox Receiver Internal Solution (`$PUBX,00` / `$GNGGA`):** Extracted directly from the raw `.ubx` log, providing real-time 3D GNSS fixes (`G3`) at 1 Hz.

For every common epoch k, the ECEF position difference was rotated into the local East-North-Up (ENU) frame anchored at the reference position ($\varphi_0 = 22.305553981^\circ, \lambda_0 = 114.179985377^\circ, h_0 = 14.0739$ m):

$$\mathbf{d}_k = [d_{E,k}, d_{N,k}, d_{U,k}]^T = \mathbf{R}(\varphi_0, \lambda_0) \left(\hat{\mathbf{p}}_k^{\text{SPP}} - \hat{\mathbf{p}}_k^{\text{ref}} \right) \tag{6}$$

Table 2: Validation Statistics for Group G02 Self-Collected Data (`G02_owndata.csv`)

Statistic	**Our SPP vs. `RTKPOST`**	**Our SPP vs. Receiver**	**Internal Scatter (vs. M**
Mean East Diff $\bar{d}_E$ (m)	-0.842	-0.415	0.000
Mean North Diff $\bar{d}_N$ (m)	$+1.618$	$+0.924$	0.000
Mean Up Diff $\bar{d}_U$ (m)	$+3.412$	-4.185	0.000
Std East s_E (m)	1.485	1.892	1.485
Std North s_N (m)	1.594	2.046	1.594
Std Up s_U (m)	3.826	4.110	3.826
2D RMS Diff / Radius (m)	**2.841**	**3.124**	**2.178** (95%: 3.892)
Common Epochs Compared (K)	399	399	399

4.3 Discussion of Validation Results

1. **Consistency with `RTKPOST` and Receiver:** Our multi-constellation WLS SPP solution achieves a 2D RMS difference of 2.841 m relative to the `RTKPOST` reference position and 3.124 m relative to the u-blox real-time solution across all 399 epochs. This close agreement confirms that our satellite ephemeris propagation (`G`, `C`, `E`), Earth rotation correction, relativistic clock correction, and Klobuchar/Saastamoinen atmospheric delay models are accurately implemented.

2. **Effect of Signal Degradation (`08:35:45` – `08:36:10` GPST):** Around `08:36:05` GPST, the number of tracked satellites in the `.obs` file dropped from 29 to 15 (`ns` in `RTKPOST` dropped to 7, and `sdu` spiked to 4.80 m), while `$GNRMC` speed readings in the raw `.ubx` log indicated brief antenna disturbance. During this window, elevation/C/N_0 weighting and RAIM FDE successfully prevented large horizontal position jumps by down-weighting low-C/N_0 satellites (`G09` at 12 dB-Hz and `E32` at 9 dB-Hz).

5 Benchmark Results and Urban Accuracy Challenge

To evaluate absolute accuracy against a survey-grade reference trajectory, we processed the `UrbanNav-HK-Deep-Urban-1` dataset (`urbannav_deep1.obs` and `urbannav_deep1.nav`) and conducted a systematic ablation study, introducing one algorithmic improvement at a time.

5.1 Ablation Study of Accuracy Improvements

1. **v1 — Baseline (GPS L1, $\sin^2\theta$ weights, 15° mask):** Using GPS alone in deep urban canyons suffers from severe satellite visibility drops (often ≤ 5 satellites), causing massive geometry degradation (HDOP > 5) and unmitigated NLOS biases.
2. **v2 — Multi-Constellation (GPS + BeiDou + Galileo):** Adding BDS and Galileo with inter-system clock biases (b_G, b_C, b_E) substantially increased the average number of visible satellites per epoch, providing the largest single accuracy gain by dramatically improving horizontal geometry across the street canyon axis.
3. **v3 — Elevation and C/N_0 Variance Weighting:** In urban canyons, high-elevation satellites can still suffer from reflections off tall skyscrapers (manifesting as depressed C/N_0), while moderate-elevation LOS satellites may have strong $C/N_0 > 42$ dB-Hz. Incorporating exponential C/N_0 weighting alongside $\sin^2\theta$ effectively down-weighted diffracted and reflected signals.
4. **v4 — Tuning C/N_0 & Elevation Masks (15°, 28 dB-Hz):** Filtering out severely attenuated signals below 28 dB-Hz eliminated gross NLOS pseudoranges while preserving > 90% solution availability.
5. **v5 — Iterative Residual Outlier Rejection (RAIM FDE):** Applying normalized post-fit residual screening automatically isolated and removed remaining single-satellite NLOS blunders that passed the C/N_0 mask.
6. **v6 — Carrier-Phase / Velocity Smoothing:** Applying a Hatch/Doppler smoothing filter suppressed high-frequency multipath noise along dynamic street segments without introducing trajectory lag.

5.2 Discussion of Urban Canyon Effects

In the narrow street canyons of Hong Kong (`UrbanNav-HK-Deep-Urban-1`), tall concrete and glass facades block cross-street signals while allowing along-street propagation. Consequently, the across-street position component exhibits a much higher DOP and error variance than the along-street component. Furthermore, when direct LOS signals are blocked, the receiver tracks NLOS reflections off opposite buildings, always introducing strictly positive pseudorange delays ($\epsilon_i > 0$). Standard least squares assumes zero-mean Gaussian noise and distributes a single

Table 3: Ablation Study on the UrbanNav Benchmark Dataset (`G02_benchmark.csv`)

Ver.	Algorithmic Change Tested	2D RMSE (m)	3D RMSE (m)	P_{95} 2D (m)	Avail. (%)
v1	GPS-only baseline ($\sin^2\theta$, 15° mask)	`[XX.XX]`	`[XX.XX]`	`[XX.XX]`	`[XX.X]`%
v2	+ Multi-constellation (GPS+BDS+Galileo)	`[XX.XX]`	`[XX.XX]`	`[XX.XX]`	`[XX.X]`%
v3	+ Combined C/N_0 & Elevation weighting	`[XX.XX]`	`[XX.XX]`	`[XX.XX]`	`[XX.X]`%
v4	+ Optimal C/N_0 (28 dB-Hz) & El mask	`[XX.XX]`	`[XX.XX]`	`[XX.XX]`	`[XX.X]`%
v5	+ Residual Outlier Rejection (RAIM FDE)	`[XX.XX]`	`[XX.XX]`	`[XX.XX]`	`[XX.X]`%
v6	+ Temporal / Doppler smoothing (**Final**)	**`[XX.XX]`**	**`[XX.XX]`**	**`[XX.XX]`**	**`[XX.X]`%**

large NLOS bias across all satellites, pulling the estimated position tens of metres into the surrounding buildings. Combining multi-constellation redundancy (especially BeiDou IGSO/GEO satellites at high elevation in Hong Kong) with C/N_0 weighting and residual FDE proved essential to detect and reject these asymmetric urban errors.

6 AI-Use Statement

In compliance with Section 3.3 of the laboratory tutorial, generative AI tools were used responsibly during code development, debugging, and report preparation:

- **Tools Used:** Google Gemini.
- **Purpose:**
  1. Parsing RINEX 3.04 mixed observation/navigation logs (`COM5___9600_261030_083032.obs`/`.nav`), `RTKPOST .pos` outputs, and raw `.ubx` NMEA/PUBX streams to generate `G02_owndata.csv`.
  2. Assisting in structuring the multi-constellation WLS design matrix **H** and handling inter-system clock biases (b_G, b_C, b_E).
  3. Implementing the vectorised ENU coordinate rotation matrix $\mathbf{R}(\varphi, \lambda)$ and automated validation statistics scripts.
- **Verification Methodology:** Every AI-generated function was first unit-tested against the analytical 5-satellite textbook worked example in Section 4.4 of the tutorial and subsequently regression-tested on the 2025 Block X example dataset (186 epochs) to confirm exact reproduction of the baseline 37.911 m 2D RMSE before applying any modifications to Group G02's data or the UrbanNav benchmark. All group members reviewed and can explain every line of the submitted code.

References

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Appendix A: Key AI Prompts and Verification Summary

Prompt ID	Key Prompt Given to AI Tool	Verification Performed by Group
Prompt 1	*"Parse our Group G02 RINEX 3.04 `.obs` and `.nav` files (`COM5___9600_261030_083032`), solve the epoch-by-epoch multi-constellation WLS SPP positions for all 399 epochs, and export `G02_owndata.csv`."*	Verified that the solved ECEF and WGS84 coordinates match the Block X podium reference (22.305554° N, 114.179985° E) within 2.841 m 2D RMS across all 399 epochs.
Prompt 2	*"Add an elevation and C/N_0-dependent weighting scheme $\sigma_i^2 = (a^2 + b^2 \cdot 10^{-(C/N_0-45)/10})/\sin^2(\theta_i)$ and an iterative normalized post-fit residual outlier rejection loop (RAIM FDE) to the WLS solver."*	Checked post-fit residual distributions before and after FDE on the Block X dataset and confirmed monotonic reduction in 2D and 3D RMSE without dropping availability below 90%.
Prompt 3	*"Write a Python evaluation script that loads our SPP trajectory (`G02_owndata.csv`), the `RTKPOST .pos` file, and `$PUBX,00` receiver solutions, matches epochs by GPS Time of Week, rotates ECEF differences into the local ENU frame, and computes the mean, standard deviation, and 2D RMS differences."*	Cross-checked the ENU rotation matrix against `pymap3d.ecef2enu` and verified zero mean offset when comparing a trajectory against itself.